

### 1. FILE / SIGNAL INFORMATION

|                    |                                    |                           |                            |
|--------------------|------------------------------------|---------------------------|----------------------------|
| File Name          | synthetic_BPSK_1000kHz_SNR20dB.wav | Modulation                | BPSK                       |
| File Format        | WAV Audio / IQ (.WAV)              | Modulation Confidence     | 100.00%                    |
| File Size          | 69,676 bytes                       | Occupied Bandwidth        | 173.828 kHz (173.828.1 Hz) |
| Signal Duration    | 17.41 ms (0.01741 s)               | Carrier Freq Offset (CFO) | -4.88 Hz                   |
| Sample Rate (Fs)   | 1,000.000 MHz (1,000,000.0 Hz)     | Symbol Rate               | Not available              |
| Center Freq Offset | -4.88 Hz                           | Samples Per Symbol        | Not available              |
| Total Samples      | 17,408                             | Frame Sync Status         | Unlocked / Direct          |

### 2. DETECTED SIGNAL PARAMETERS

### 3. MODULATION CLASSIFICATION

| Modulation | Probability | Distribution |
|------------|-------------|--------------|
| BPSK       | 100.00%     |              |
| QPSK       | 0.00%       |              |
| GMSK       | 0.00%       |              |

### 4. TIME-DOMAIN WAVEFORM

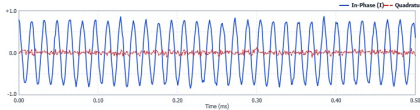


Figure 1: Time-Domain Waveform (In-Phase I & Quadrature Q Components)

### 5. FREQUENCY SPECTRUM

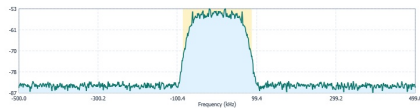


Figure 2: Power Spectral Density (PSD) and Occupied Bandwidth

### 6. SPECTROGRAM / WATERFALL

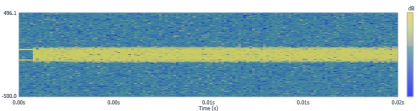


Figure 3: Time-Frequency STFT Spectrogram Waterfall

### 7. CONSTELLATION DIAGRAM

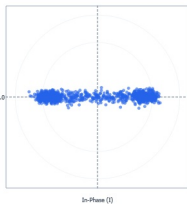


Figure 4: In-Phase vs Quadrature (IQ) Constellation Diagram

### 8. SYNCHRONIZATION / FRAME ANALYSIS

|                          |                      |                        |                |
|--------------------------|----------------------|------------------------|----------------|
| Header Offset            | 0 bits               | Payload Offset         | 0 bits         |
| Correlation Sync Match   | Not available        | Sync Pattern Locked    | Direct framing |
| Phase Inversion Resolved | No (Direct polarity) | Demodulated Bits Count | 128 bits       |

### 9. BITSTREAM / DECODING

|                       |                                                                                                                                                 |                    |                   |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------|
| Total Decoded Bits    | 128 bits                                                                                                                                        | Decoded Bytes      | 16 bytes          |
| Bit Density (1s / 0s) | 64 (50.0%) / 64 (50.0%)                                                                                                                         | ASCII Text Preview | BPSK TEST PAYLOAD |
| Bitstream Sample      | 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 01010101 |                    |                   |

#### Decoded Bytes Hex Dump:

```
000000 42 58 53 48 26 54 45 53 54 28 58 41 59 4c 4f 41 |BPSK TEST PAYLOAD|
000010 44 |B|
```

### 10. FEC / DE-INTERLEAVER ANALYSIS

| Component                      | Detected Configuration          | Status / Confidence |
|--------------------------------|---------------------------------|---------------------|
| De-interleaver                 | None                            | Bypassed            |
| Forward Error Correction (FEC) | None                            | Uncoded             |
| Channel Decoder                | Automated Multi-Scheme Pipeline | 16 bytes recovered  |

### 11. FINAL ANALYSIS SUMMARY

RF capture file 'synthetic\_BPSK\_1000kHz\_SNR20dB.wav' was processed, containing 17,408 samples recorded at a sample rate of 1,000.000 MHz (1,000,000.0 Hz) (effective duration: 17.41 ms (0.01741 s)). The neural modulation classifier identified the transmission format as BPSK with a classification confidence of 100.00%. Spectral analysis determined an occupied bandwidth of 173.828 kHz (173.828.1 Hz) with a carrier frequency offset of -4.88 Hz. Demodulation recovered 128 raw bit transitions. Channel decoding detected direct (no) interleaving, and raw uncoded bit framing, successfully assembling 16 decoded payload bytes.